RelationalRAG-Tables: A Learned Join Verifier for QA over Tables and Knowledge Graphs

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Abstract

Questions that span relational tables and knowledge graphs (KGs) depend on a valid join between sources. When identifiers differ in format, hierarchy level, or vocabulary, exact matching can miss a valid link, while accepting a plausible but incorrect link can produce a wrong answer. We propose **RelationalRAG-Tables**, which uses a small learned verifier to guide this decision. Our planner checks each candidate join, tries another candidate after rejection, and abstains when uncertainty is high. We study how this use of existing matching methods affects question answering (QA), supported by two conditional theorems on recall gains and the utility of abstention. Our evaluation spans **20** synthetic seeds, constructed medical records linked to Wikidata, BIRD value linking and query generation, and HybridQA evidence retrieval. On BIRD with corrected query-alias resolution, our verifier recovers drifted values at five-seed recall **0.91** versus **0.00** for exact matching. On the constructed table+KG benchmark, gating reduces silent wrong answers, incorrect answers returned without abstention, from 14.0% to 0.7%. The benefit depends on how the match is used. Direct join gating reduces wrong answers, while SQL hints remain limited by selecting the correct database column. Hierarchy rules, lexical overlap, and tuned rule cascades tie or beat learned verification on the data they suit. We release a dependency-light implementation, benchmark generator, evaluation harness, and label-audit package.*

*Code and data: https://github.com/lxbachcs/relationalrag-tables

Keywords: data integration, entity matching, record linkage, knowledge graphs, retrieval-augmented generation, value linking, uncertainty calibration, selective prediction, text-to-SQL

1 Introduction

Consider the question "*which instrument does account A-1001 hold, and who regulates its asset class?*" The account's holdings may reside in a relational table, while a *knowledge graph* (KG) represents the instrument's asset class and its relation to a regulator. To answer, we must join the table row to the right graph node before following those relations. Retrieval-augmented generation (RAG) typically supplies textual evidence to a generator (Lewis et al., 2020), but flattening rows and graph relations into text can obscure this join. Even a correctly executed query returns the wrong answer if it starts from the wrong entity.

The difficulty is that a valid join need not connect identical strings. For example, `ACC-1001` in a table may correspond to `ACX1001` in the graph, and a specific disease code may link to a broader category. We call differences in format, hierarchy level, or vocabulary *identifier drift*. Exact matching can reject these valid links, while a similar-looking neighboring entity can attract an incorrect match. Cell-aware encoders (Herzig et al., 2020), text-to-SQL systems that translate questions into Structured Query Language (SQL) (Wang et al., 2020; H. Li et al., 2023, 2024; Wang et al., 2025), and knowledge-base question answering (KBQA) pipelines (Sun et al., 2020) address parts of the task. A tool-using planner (Yao et al., 2023) must still decide which cross-source links to trust.

We introduce **RelationalRAG-Tables**, whose *learned join verifier* estimates whether a table key and a candidate graph node satisfy the task's linking rule. Our planner uses this match score and its uncertainty to commit a join, backtrack to another candidate, or abstain. If the requested entity is absent, for instance, rejecting a plausible neighbor can prevent an incorrect answer. We aim to reduce these *silent wrong answers*, incorrect answers returned without abstention, while preserving useful coverage, the fraction of questions we answer.

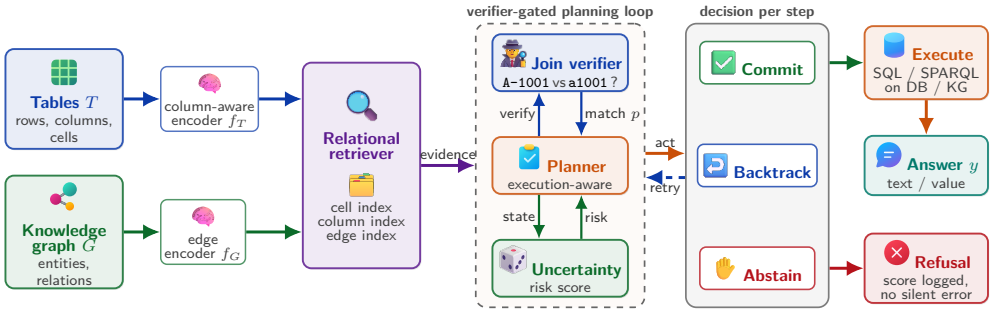


Fig. 1: Our retrieval and verification pipeline. The planner checks candidate joins and uncertainty before committing, backtracking, or abstaining.

In Figure 1, encoders f_T, f_G map tables T and graph G into cell, column, and edge indices. The dashed loop checks match probability p and uncertainty before execution, backtracking to another candidate or abstaining with a logged refusal. Accepted joins support SQL or SPARQL, a graph query language, to produce answer y. DB denotes database. We thus check retrieved candidates before their errors propagate into answers.

Verification can prevent a wrong answer, but rejecting a valid join also sacrifices coverage. We study this balance in cross-source question answering (QA). Our contributions are:

1. **A verifier-guided planner and conditional analysis.** We integrate existing matching methods into a planner that can commit, backtrack, or abstain. Two theorems characterize gains in verification recall, the fraction of true matches recovered, and in the utility of abstention (Sections 3 and 7).
2. **A reproducible implementation and controlled evaluation.** We release a dependency-light system and benchmark generator, and evaluate 20 synthetic seeds. We measure calibration, agreement between probabilities and match frequencies, and selective prediction, deferring uncertain cases, alongside empirical checks of the theory (Sections 4 and 5).
3. **Evidence across matching tasks and complete QA pipelines.** We evaluate constructed medical records linked to Wikidata, corrected BIRD value linking, BIRD query generation, and HybridQA evidence retrieval. Comparisons with rules, tuned cascades, and lexical overlap identify where learned verification helps and where simpler methods suffice (Section 6).

Our results connect matching quality to the decision it controls. A planner can reject an uncertain join directly, while a SQL generator must identify the relevant column and use a match hint correctly. This distinction explains why recovering identifiers alone need not improve the final answer.

2 Related Work

Table and knowledge-graph question answering.

Table structure is modeled by column-aware Transformers such as TaPas (Herzig et al., 2020). Semantic parsers, including RAT-SQL (Wang et al., 2020), RESDSQL (H. Li et al., 2023), CodeS (H. Li et al., 2024), and multi-agent extensions (Wang et al., 2025) translate questions into SQL over a database schema. KBQA includes skeleton parsers (Sun et al., 2020), while HybridQA (Chen et al., 2020) combines tables with linked passages. Hybrid RAG systems that route questions over a KG and text, such as SBV-LawGraph (Phan et al., 2026), take the entity links as given. A valid SQL join clause leaves drifting identifiers unresolved, while HybridQA supplies cell-to-passage links. We evaluate identity matching and evidence retrieval separately to account for their different matching signals.

Tool-using RAG, selective prediction, and uncertainty.

Building on generation conditioned on retrieved text in RAG (Lewis et al., 2020) and tool calls in ReAct (Yao et al., 2023), we treat join feasibility as a planner decision. LTM-RAG scores the trustworthiness of retrieved sources over time (T-H-C. Le et al., 2026), whereas we score the join itself. *Selective prediction* trades *coverage*, the fraction of cases answered, for lower *risk*, the error rate among those answers (Geifman & El-Yaniv, 2017). Deep ensembles combine independently trained models to estimate predictive uncertainty (Lakshminarayanan et al., 2017; Fort et al., 2019). Conformal methods use held-out calibration data to control risk under stated assumptions (Angelopoulos & Bates, 2023). We use ensemble uncertainty during planning and evaluate the resulting coverage–risk tradeoff empirically.

Entity matching and value/schema linking.

Identifying corresponding entities across sources is the task of entity matching, also called record linkage (Konda et al., 2016; Christophides et al., 2020), for which learned matchers such as DeepMatcher and Ditto are established approaches (Mudgal et al., 2018; Y. Li et al., 2020). Related tasks include *value linking*, which maps a question phrase to a stored cell value (Lin et al., 2020), and *schema linking*, which identifies the relevant tables and columns (Wang et al., 2020; H. Li et al., 2023, 2024). Table-to-knowledge-base alignment also relies on matching across representations (Efthymiou et al., 2017; Sung et al., 2020).

Structural join discovery identifies which tables can be joined. We examine retrieved key–node pairs and their effects on planning, comparing with Fellegi–Sunter probabilistic linkage (Fellegi & Sunter, 1969), a Ditto-style cross-encoder that jointly encodes each candidate pair (Y. Li et al., 2020; Mudgal et al., 2018), and rule cascades tuned on training data.

3 Method

A candidate may be the most likely match and still leave too much uncertainty to answer. We handle this distinction with separate acceptance and abstention thresholds, first formalizing the join decision and its costs, then describing how our planner acts on the result.

Definition 1 (Heterogeneous QA)**.** *A corpus is $\mathcal{D} = (T, G)$ where $T = \{T_1, \ldots, T_M\}$ is a set of relational tables, each T_m a set of rows over typed columns C_m (M is the number of tables), and $G = (N, E, R)$ is a knowledge graph with node set N, edges $E \subseteq N \times R \times N$, and relation vocabulary R. A question q has a (latent) execution trace $\pi^\star = (o_1, \ldots, o_L)$ of length L, where each o_ℓ is a typed relational operation (`select`, `filter`, `join`, `traverse`, `return`, or `abstain`) and a gold answer $y^\star$ (possibly the distinguished value `ABSTAIN`).*

Our controller builds a predicted trace π from its execution history. The executor $\mathcal{E}$ maps each trace prefix to intermediate results drawn from the corpus. We use two metrics. *Exact match* (EM) is $\mathbb{I}[\mathcal{E}(\pi) = y^\star]$ after answer normalization, where $\mathbb{I}$ is the indicator function, and *faithfulness* asks whether the returned value appears in

retrieved cells or edges or is the deterministic result of a traversal. The difficult operation is `join`. Given a key cell value ρ and a candidate node ν, we check entity identity in the synthetic setting or a valid code-to-category mapping in the medical setting. Thus, different strings can form a valid join, as when a specific disease code maps to its broader category. Our baseline uses substring matching, and our verifier learns a match probability.

3.1 Theoretical Properties

We analyze the utility of abstention, recall under drift, and ensemble probability quality.

Theorem 1 (Abstention reduces expected loss under asymmetric costs)**.** *Consider a population of join decisions, scored one decision at a time: a correct accepted join scores* $+1$*, a wrong accepted join* $-\lambda$ $(\lambda > 0)$*, and an abstention* 0*. Assume join decisions are independent and the verifier's errors are independent of retrieval. For a verifier with false-positive rate (FPR)* α *(accepting a non-match) and false-negative rate (FNR)* β *(rejecting a match) on a set of* n *join decisions with match prevalence* π_+ *(the fraction of true matches), the expected per-decision utility under gated abstention is*

$$U_{\text{gate}} = \pi_+(1-\beta) - \lambda\,(1-\pi_+)\alpha,$$

while the ungated planner achieves $U_{\text{no}} = \pi_+ - \lambda(1-\pi_+)$*. Gating improves expected utility whenever* $\lambda > \pi_+\beta\,/\,[(1-\pi_+)(1-\alpha)]$*, i.e., when wrong joins are sufficiently costly relative to the verifier's error rates.*

Proof. The ungated planner commits every candidate: it accepts correctly on the $\pi_+ n$ true matches and accepts a wrong join on the $(1-\pi_+)n$ non-matches, giving $U_{\text{no}} = \pi_+ - \lambda(1-\pi_+)$. Under gating, the verifier accepts true positives at rate $(1-\beta)$ and false positives at rate α, yielding $U_{\text{gate}} = \pi_+(1-\beta) - \lambda(1-\pi_+)\alpha$. The difference $U_{\text{gate}} - U_{\text{no}} = -\pi_+\beta + \lambda(1-\pi_+)(1-\alpha)$ is positive when $\lambda > \pi_+\beta\,/\,[(1-\pi_+)(1-\alpha)]$, which is the stated bound. □

Corollary 1 (Wrong-join and abstention rates under gating)**.** *Under the assumptions of Theorem 1, gating reduces the rate of wrong joins from* $(1-\pi_+)$ *to* $(1-\pi_+)\alpha$ *and introduces an abstention rate of* $\pi_+\beta + (1-\pi_+)(1-\alpha)$*. The guarantee is conditional on* α*: it bounds wrong admissions only as tightly as the verifier's measured false-positive rate.*

Rejecting a wrong account avoids an error penalty, but rejecting the correct account forfeits an answer. We measure the critical penalty $\lambda^\star = \pi_+\beta/[(1-\pi_+)(1-\alpha)]$ in Section 7. Synthetically, $\lambda^\star = 0.39$ predicts a benefit for $\lambda \geq 1$, agreeing with measured utility in all 20 seeds.

Theorem 2 (Regime separation: when learned verification improves over exact matching)**.** *Let* $\delta \in [0,1]$ *be the fraction of join pairs whose identifiers are format-drifted (not string-equal to the gold). An exact matcher has recall* $\text{Rec}_{\text{ex}} = 1-\delta$ *on matches. A learned verifier with recall* r *on drifted pairs and recall* r_0 *on clean pairs achieves* $\text{Rec}_{\text{learn}} = (1-\delta)\,r_0 + \delta\,r$*. The learned verifier strictly improves recall whenever* $\delta > 0$ *and* $r > (1-r_0)(1-\delta)/\delta$*, which is satisfied for any* $r > 0$ *when* $r_0 = 1$ *(no clean-pair loss) and* $\delta > 0$*.*

Proof. The improvement $\mathrm{Rec}_{\mathrm{learn}} - \mathrm{Rec}_{\mathrm{ex}} = (1-\delta)(r_0 - 1) + \delta\, r$. When $r_0 = 1$ this equals $\delta\, r > 0$ for any $\delta, r > 0$. Otherwise, the condition is $\delta\, r > (1-\delta)(1-r_0)$, yielding the stated bound. □

The recall gain balances recovered drifted matches against lost clean matches. We measure its inputs on each corpus's gold join pairs: over 20 seeds $\delta = 0.553_{\pm 0.021}$, $r = 0.926_{\pm 0.047}$, $r_0 = 0.9996$, giving a predicted gain of $0.512_{\pm 0.032}$ recall points against an observed join-task EM gain of $0.515_{\pm 0.025}$ (mean absolute error 0.013, Section 7).

Lemma 1 (Ensemble averaging does not worsen the Brier score)**.** *For a K-member ensemble with member predictions $\hat{p}_1, \ldots, \hat{p}_K$ and mean $\bar{p} = K^{-1} \sum_k \hat{p}_k$, the Brier score (mean squared probability error against a binary label $y \in \{0, 1\}$) of $\bar{p}$ is at most the mean Brier score of the members, by Jensen's inequality applied to the convex map $p \mapsto (p - y)^2$. This bound does not guarantee better calibration. We assess calibration separately through expected calibration error (ECE), the bin-weighted gap between predicted probabilities and empirical match frequencies (Section 5).*

3.2 System Design

We implement these decisions in the retrieval–planning loop of Figure 1.

3.2.1 Encoding and Relational Retriever

We encode each cell value together with a column-type embedding $\phi_{\mathrm{col}}(c_j)$, where $c_j \in C_m$ is its column, and encode graph triples similarly. The deterministic feature-hash embeddings map features to fixed vector coordinates without a pretrained model, isolating the verifier from memorized facts. Column-aware Transformers are an extension (Herzig et al., 2020). Cells, column headers, and edges populate separate indices $\mathcal{I}_C$, $\mathcal{I}_H$, and $\mathcal{I}_E$. For question q and candidate x, we score $s(q, x) = \langle f_q(q), f_x(x) \rangle / \gamma$, where f_q, f_x are their encoders, $\langle \cdot, \cdot \rangle$ is a dot product, and $\gamma > 0$ is a score scale. We return the k highest-scoring candidates from each view.

3.2.2 Execution-Aware Planner

At each step, our planner emits one of the typed operations defined above. For a `join`, the planner queries verifier V for a match probability and uncertainty head U for a confidence signal (Algorithm 1). Candidates are considered in retrieval order. If the match probability falls below the acceptance threshold τ_v, we backtrack. Otherwise, uncertainty above the budget τ_u ends the question with abstention. Only a candidate passing both tests is committed, and exhausting the pool also triggers abstention. In Algorithm 1, h_ℓ is the execution history after step ℓ, $\hat{p}$ the match probability, and $L_{\max}$ the step limit. We use a deterministic, rule-guided controller and discuss learned planning under Limitations and Future Work.

3.2.3 Learned Join Verifier

Given (ρ, ν), our verifier predicts $\hat{p} = \sigma(g_\phi(\rho, \nu))$, where g_ϕ is a learned score with parameters ϕ and σ is the logistic sigmoid. The reference implementation builds an explicit feature vector (exact match, containment, digit agreement, n-gram Jaccard,

Algorithm 1: RelationalRAG-Tables planner loop

Input: question q, corpus (T, G), retriever $\mathcal{R}$, verifier V, uncertainty head U, thresholds τ_v, τ_u, step budget $L_{\max}$

$h_0 \leftarrow \text{encode}(q)$, $\pi \leftarrow [\,]$;
for $\ell = 1, \ldots, L_{\max}$ **do**
  $o_\ell \leftarrow$ next typed operation for q given history $h_{\ell-1}$;
  if o_ℓ.type = *join* **then**
    committed $\leftarrow$ **false**;
    foreach *candidate node* ν *in retriever order* **do**
      $\hat{p} \leftarrow V(o_\ell.\text{key}, \nu)$;
      if $\hat{p} < \tau_v$ **then continue** (reject, backtrack);
      if $U.\text{uncertainty}(o_\ell.\text{key}, \nu) > \tau_u$ **then return** `ABSTAIN`;
      commit ν, *committed* $\leftarrow$ **true**, **break**;
    if $\neg$*committed* **then return** `ABSTAIN`;
  $h_\ell \leftarrow \mathcal{E}(o_\ell, h_{\ell-1})$, $\pi \leftarrow \pi \cup \{o_\ell\}$;
  if o_ℓ.type = *return* **then return** h_ℓ;

edit-ratio, length ratio, prefix overlap, embedding cosine, token Jaccard) and trains a small multilayer perceptron (MLP) on matches and hard negatives, plausible but incorrect pairs. Its cross-entropy loss penalizes incorrect match probabilities. Jaccard similarity is set intersection over union, edit-ratio is normalized edit similarity, cosine is a normalized vector dot product, and n-grams are length-n character sequences. The shared digits in `ACC-1001` and `ACX1001`, for example, remain informative despite the changed prefix. Our verifier learns how much weight to give this agreement alongside the other cues. We assess discrimination by the area under the receiver operating characteristic curve (AUROC), which measures ranking across thresholds, with 0.5 indicating chance performance.

3.2.4 Uncertainty via Deep Ensembles

We train K independently seeded verifiers on bootstrap resamples (training pairs sampled with replacement) (Lakshminarayanan et al., 2017) with diverse hidden widths. The ensemble exposes mean probability $\bar{p}$ (gating joins) and predictive uncertainty $H(\bar{p}) = -\bar{p}\log\bar{p} - (1-\bar{p})\log(1-\bar{p})$ (gating abstention). We use $2\,\text{Var}_k(p_k)$, twice the variance across member probabilities p_k, to diagnose model disagreement. Largest near $\bar{p} = 0.5$, where match and non-match are equally plausible, entropy H reflects both model uncertainty and ambiguity in the data. We use it to rank and defer uncertain cases through selective prediction (Section 5.4).

4 Implementation and Benchmark

Our synthetic core runs from a seed in minutes on a laptop's central processing unit (CPU), requiring no pretrained model, graphics processing unit (GPU), external search library, or network access. We use deterministic feature-hash encoders, in-memory JavaScript Object Notation (JSON) indices, a scikit-learn MLP, and the planner in Algorithm 1. Real-data studies add sentence embeddings and, for SQL, a

quantized 7–14B coder, where B denotes billion parameters. The reproduction guide maps results to commands and outputs, including the corrected BIRD driver and snapshot.

Synthetic generator

We generate a seeded corpus of an `accounts` table joined to a financial-instruments knowledge graph (instrument → asset class → regulator), with five task families: single-table lookup, table→KG join, multi-hop traverse (two to three hops), bad-join backtrack (a wrong neighboring node precedes the correct one), and abstain ("ghost" accounts absent from the graph). Accounts reserved for verifier training and evaluation form a disjoint split and never appear in QA questions.

Identifier drift

A configurable fraction of KG node identifiers are *format-drifted* relative to the table key by mutating the alphabetic prefix while *preserving the numeric key* (table `ACC-1001` versus node `account:ACX1001`). Substring and exact matching then fail, although a verifier that weights digit agreement can recover the join. Hierarchy drift poses a different challenge, mapping a specific medical code to a broader category. Because our distractors are *real neighboring entities*, accepting one produces a concrete wrong answer.

Real table+KG benchmark and its end-to-end form

We build a constructed clinical-records table joined to a disease KG from 71,704 billable codes in the International Classification of Diseases, Tenth Revision, Clinical Modification (ICD-10-CM), and Wikidata disease entities carrying an ICD-10 code (property P494). A record's specific code can map to a broader graph category (Section 6.1). For end-to-end QA (Section 6.2), we introduce hierarchy drift, optical character recognition (OCR)-style corruption, or vocabulary drift through a legacy ninth-revision ICD-9 code. 12% of test records are "ghosts" with no KG node. We extend the graph with live-fetched Wikidata relations for specialty, anatomical location, drug, subclass-of, and parents' specialties (2,126 triples). We instantiate the synthetic generator's five question families to obtain 518 questions per seed.

Label protocol

Our labeling rules follow the join semantics of each dataset. For BIRD, we take a literal from a gold-SQL equality condition and confirm it in the alias-resolved table and column. The case is drifted if the normalized literal is absent from the normalized question. For ICD, the gold node carries the category of the record's billable code, and we record the drift subtype during construction. Two annotators independently audited 100 items per dataset, then adjudicated 14 BIRD and 5 ICD rows. Measured by Cohen's κ, chance-adjusted agreement was 0.81 for BIRD gold correctness and 1.00 for drift, with adjudicated error rates of 0.12 and 0.00. All 12 gold errors arose from choosing a same-named column in the wrong table. On ICD, every gold node was accepted (κ undefined because judgments had no variance). Drift agreement was 1.00, but adjudication found five no-op OCR edits that remained hierarchy cases.

Propagation of the audit corrections

Correcting SQL-alias resolution changes 89 BIRD table assignments. We rerun value linking on the corrected pools and report the comparison in Section 6, using recall@k, the fraction of cases with the gold candidate among the top k. The ICD audit changes only subtype metadata. We retain aggregate results because codes, gold nodes, features, and predictions are unchanged, and omit the invalid subtype breakdown.

Sensitivity to label noise

To test sensitivity to labeling errors, we corrupt a fraction η of training gold assignments while keeping the test split clean. Half of the corrupted ICD assignments use same-category siblings. At high noise, false rejection increases while false admission remains rare. The supplementary material reports recall and error rates across noise levels, including η=0.2. This sweep tests the specified assignment errors, leaving other error mechanisms for further study.

Verifier split

Independently, we generate training and held-out (key, node) pairs under *graded corruption*. We sample the number of character edits and decrease the probability that the corrupted identifier denotes the same entity as edit distance grows, reaching a $\sim$50/50 boundary at two edits. The same degree of surface similarity can then describe either a match or a non-match, creating ambiguous pairs on which ensemble members can disagree.

5 Experiments

Our evaluation follows the path from recovering a drifted join to deciding whether to answer a question. We begin with a controlled synthetic corpus, where we can examine verification and uncertainty separately, then test their effects on real identifiers and complete QA pipelines.

5.1 Setup

Unless noted, we use a corpus of 600 accounts (seed 7), with 300 reserved for the verifier, 60 for development, 36 ghost accounts, and 342 format-drifted nodes at drift fraction 0.6. This yields 1,392 QA questions, 4,000 verifier-training pairs, and 1,500 held-out verifier-evaluation pairs. Holding the planner fixed, we compare five gates. **Exact-match** uses substring matching, **No verifier** commits the first retrieved candidate, **Learned** uses a single MLP, **Deep ensemble** combines $K = 5$ members, and **Oracle** uses gold join labels as an upper bound. With the join-acceptance threshold fixed at $\tau_v = 0.5$, we vary corpus and verifier seeds together over 20 runs and report means with standard deviation (sd) as a gray subscript, t-based 95% confidence intervals (CIs), and bootstrap curve bands. The task-family bars show seed 7. Designated real-data studies use five seeds and paired bootstrap tests with 10,000 resamples.

5.2 End-to-end QA

In Figure 2, Single denotes lookup, Join table-to-KG linking, Multi multi-hop traversal, Back rejection of a wrong candidate, and Abstain refusal of an unanswerable question. Our exact-match baseline reaches only $0.580_{\pm 0.012}$ aggregate EM because it misses format-drifted identifiers (join, multi-hop, and backtrack average 0.45, reflecting the clean fraction of the evaluated questions) and over-abstains, refusing 620 of 1,392 questions on average. Our learned verifier and ensemble recover the drifted joins (0.95 to 0.96 on join and multi-hop) and reach $0.854_{\pm 0.026}$ and $0.862_{\pm 0.016}$ EM, approaching the oracle ceiling of 1.0 on the easy task families.

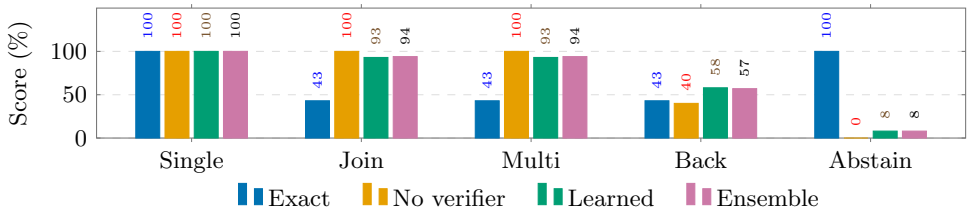


Fig. 2: Synthetic EM by task family (seed 7, 1,392 questions, configured drift 60%).

Why aggregate EM under-rewards verification

Similar aggregate EM ($0.848_{\pm 0.006}$ ungated versus $0.862_{\pm 0.016}$ ensemble) hides different errors. Committing every candidate gives the ungated planner 1.0 on single/join/multi-hop questions, but wrong answers for all 36 absent accounts and two thirds of bad-join questions. Our gate trades some join recall for rejection. The Back and Abstain bars show that improved backtracking alone does not ensure reliable refusal.

5.3 Held-out verifier discrimination

We separate pair classification from downstream QA in Table 1. Accuracy (Acc) is the fraction of correct pair labels, precision (Prec) is the match fraction among accepted pairs, recall (Rec) is the accepted fraction of true matches, and F1 is their harmonic mean. High precision hides a recall weakness in substring matching, which misses most true matches among the 1,500 held-out pairs per seed (recall 0.37).

Learned verification recovers missed joins. Recall rises from 0.37 to 0.79 and AUROC reaches $0.815_{\pm 0.009}$. Our ensemble reaches AUROC $0.820_{\pm 0.008}$ and lowers the Brier score from 0.37 to 0.17, improving probability quality as well as recall.

Table 1: Pair classification on 1,500 held-out graded-corruption pairs per seed. Means over 20 seeds. Gray subscripts give sd and arrows indicate the preferred direction.

Verifier	Acc ↑	Prec ↑	Rec ↑	F1 ↑	AUROC ↑	Brier ↓
Exact match	$0.633_{\pm 0.012}$	$0.961_{\pm 0.012}$	$0.365_{\pm 0.014}$	$0.529_{\pm 0.015}$	$0.673_{\pm 0.007}$	$0.366_{\pm 0.012}$
Learned	$0.739_{\pm 0.015}$	$0.758_{\pm 0.023}$	$0.789_{\pm 0.034}$	$0.773_{\pm 0.016}$	$0.815_{\pm 0.009}$	$0.174_{\pm 0.005}$
Ensemble	$0.741_{\pm 0.010}$	$0.760_{\pm 0.021}$	$0.793_{\pm 0.038}$	$0.775_{\pm 0.014}$	$0.820_{\pm 0.008}$	$0.171_{\pm 0.004}$

5.4 Deep-ensemble selective prediction

Figure 3 plots accuracy (1 − risk) as coverage falls in 1% steps, with shaded 95% CIs over seeds. Moving right defers more cases, and higher curves mean lower risk. Pair curves evaluate match labels, while the system curve evaluates QA answers, including easy lookups. The bracket marks the 10.8-point pair-accuracy gap at full coverage. Area under the risk–coverage curve (AURC) integrates the complement of the plotted accuracy. Table 2 separates calibration from selective accuracy, with maximum calibration error (MCE) giving the largest bin-wise probability–frequency gap. Deferring half of pairs raises accuracy from 0.74 to 0.86 (ensemble AURC $0.132_{\pm 0.006}$ versus exact $0.229_{\pm 0.011}$). Ensemble ECE is $0.026_{\pm 0.010}$, compared with $0.040_{\pm 0.016}$ for a single verifier and 0.37 for exact matching. BIRD ECE is 0.001. Non-abstained answers are faithful to retrieved evidence in 0.94 of cases.

We select thresholds on a calibration half and evaluate on the disjoint test half. Maximizing calibration F1 gives τ_v of $0.43_{\pm 0.06}$ (range 0.35–0.55), with test F1 $0.779_{\pm 0.011}$ versus $0.772_{\pm 0.016}$ at our fixed τ_v=0.5. F1 stays within $[0.75, 0.77]$ on synthetic pairs and $[0.79, 0.80]$ on ICD pairs for $\tau_v \in [0.4, 0.6]$. Split-conformal selection instead targets a specified join-risk α_{risk} (Geifman & El-Yaniv, 2017; Angelopoulos & Bates, 2023), distinct from the FPR α in Theorem 1. Test risk slightly exceeds each target on average (Table 2, right). Here cov. is test coverage and held counts seeds meeting the target. These ensemble-only rows are separate from the left-hand calibration results over 20 synthetic seeds and a BIRD diagnostic.

Deferring uncertain answers improves accuracy and utility. QA EM rises from 0.88 at full coverage to 0.95 at 70% and 1.00 at 50% coverage (AURC $0.031_{\pm 0.007}$). When a wrong answer costs 5× the reward for a correct one, utility rises from $0.270_{\pm 0.070}$ to $0.600_{\pm 0.066}$ at optimal coverage $0.62_{\pm 0.08}$.

A single model and the ensemble both reach pair accuracy 0.86 at 50% coverage, giving similar selective ranking. The ensemble instead improves calibration. Recalibrating a single model by temperature scaling of logits or a monotone isotonic probability fit gives ECE 0.039 and 0.043, respectively, compared with 0.024 for the ensemble. Correlation among ensemble members limits the benefit of adding models (Fort et al., 2019; Lakshminarayanan et al., 2017), which we examine next.

Table 2: Calibration and selective prediction. Left: calibration metrics. Right: a separate ensemble risk-target study. Bottom: accuracy at fixed coverage for pairs and QA answers.

Calibration and ranking (pairs)					*Ensemble threshold at target risk*			
Split	Verifier	ECE ↓	MCE ↓	AURC ↓	α_{risk}	cov.	risk	held
Synthetic	Exact	$0.366_{\pm 0.012}$	$0.366_{\pm 0.012}$	$0.229_{\pm 0.011}$	0.02	$0.189_{\pm 0.067}$	$0.029_{\pm 0.014}$	4/20
Synthetic	Learned	$0.040_{\pm 0.016}$	$0.082_{\pm 0.033}$	$0.139_{\pm 0.008}$	0.05	$0.273_{\pm 0.036}$	$0.057_{\pm 0.022}$	8/20
Synthetic	**Ensemble**	$0.026_{\pm 0.010}$	$0.059_{\pm 0.028}$	$0.132_{\pm 0.006}$	0.1	$0.387_{\pm 0.052}$	$0.105_{\pm 0.027}$	10/20
BIRD	**Ensemble**	**0.001**	0.053	—				

Accuracy at coverage	1.00	0.90	0.80	0.70	0.50
Pairs (ensemble)	$0.741_{\pm 0.010}$	$0.770_{\pm 0.012}$	$0.788_{\pm 0.010}$	$0.808_{\pm 0.009}$	$0.858_{\pm 0.009}$
System (answered)	$0.878_{\pm 0.012}$	$0.900_{\pm 0.014}$	$0.913_{\pm 0.028}$	$0.946_{\pm 0.035}$	$0.999_{\pm 0.003}$

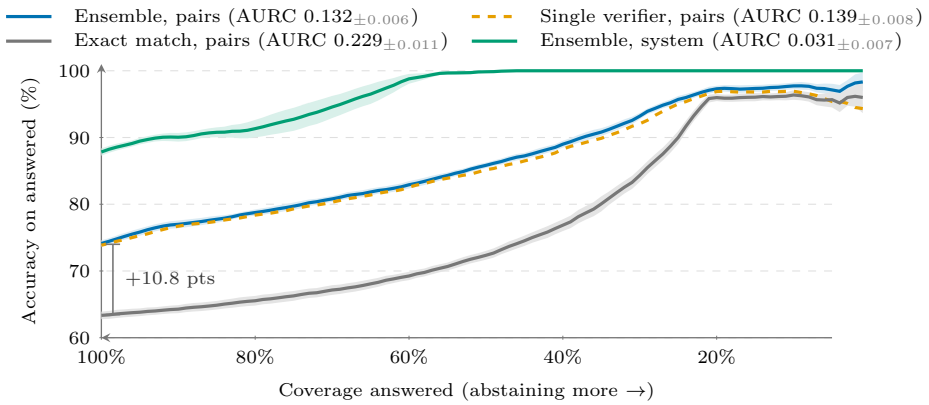


Fig. 3: Accuracy versus coverage for held-out pairs and QA answers over 20 seeds. Shaded bands show 95% CIs where available.

5.5 Ensemble size and feature ablation

Adding members beyond $K = 3$ yields little further gain. Across 20 seeds with nested ensemble members, AUROC is 0.815, 0.820, 0.820 and EM is 0.854, 0.862, 0.862 for $K \in \{1, 3, 5\}$. A smaller ensemble may therefore suffice for ranking alone. To identify which cues support the result, we retrain on single-feature subsets. Digit agreement reaches AUROC 0.70, edit-ratio 0.76, prefix overlap 0.67, and embedding only 0.52, compared with 0.82 for the full combination. The supplementary material gives per-K selective-prediction results and the full feature ablation.

5.6 Cost and robustness checks

Per-question latency is ∼11 ms, with sub-millisecond verification. On an undrifted corpus, exact matching achieves EM 1.0. Retrieval failures are more revealing. When we drop the gold candidate with probability p on the 264 bad-join questions, the ungated wrong-answer rate stays at 0.60 for every p. Our ensemble converts failures into abstentions, whose rate rises from 0.02 at p=0 to 0.29 at p=1, and limits wrong answers to 0.46 at p=1. This is a 23% relative reduction in silent errors, with the benefit widening as retrieval quality falls.

6 Real-Data Validation

We test component matching and end-to-end QA on medical-code joins, BIRD value linking and text-to-SQL, and HybridQA evidence retrieval.

Evidence map.

Table 3 distinguishes component matching from system outcomes. Even after a linker recovers a value in a known column, a SQL generator must identify that column and use the value correctly. Execution accuracy (EX) measures whether predicted SQL returns the gold query result, and gains in EX are percentage points. In the table,

"sig." means statistically significant, "n.s." means not significant, and p is a bootstrap significance value.

Table 3: Evidence map distinguishing component matching from system outcomes. Corrected BIRD value linking and original-extractor SQL generations are reported separately.

Claim (dataset)	Result	Level	Status
Drifted joins (synth.)	AUROC 0.67→0.82, EM 0.58→0.86	pairs, system	20 seeds
Selective pred. (synth.)	ECE $0.026_{\pm 0.010}$, AURC 0.23→0.13	pairs, system	20 seeds
Theorems 1–2 (synth.)	pred. $0.512_{\pm 0.032}$, obs. $0.515_{\pm 0.025}$	theory	20 seeds
Value linking (BIRD)	recall@1 0.00→0.91 on drift	component	5 seeds, corrected rerun
Hierarchy join (ICD-10)	0.75, ties prefix rule	component	5 seeds, p=0.59
Cross-vocab. (ICD-9→10)	rules 0.01, learned 0.90	component	single seed
Heterogeneous (ICD)	0.75 vs. rule 0.26, tuned cascade ties	component	5 seeds
Table+KG QA (ICD)	wrong 14.0%→0.7%, EM 0.860 vs. 0.839	**system**	sig. 5/5, EM n.s.
End-to-end EX (BIRD)	oracle hints +3.1, verifier hints $\leq$+0.4	system	full dev, bootstrap
Negative (HybridQA)	lexical 0.25 > verifier 0.09	component	single seed

6.1 Real table+KG join: ICD-10 + Wikidata

Our ICD-10 benchmark links billable codes to 590 Wikidata disease entities carrying an ICD-10 code (property P494). A table record may carry the specific code `R509`, while a graph node carries its broader category `R50`. Although these strings differ, the code-to-category join is valid. Among the 590 records, 164 (28%) are drifted, and for 159 of them no KG node carries the record's exact code.

Evaluation protocol.

We rank all 590 Wikidata disease nodes for ICD and the true column domain for BIRD. We shuffle candidates before scoring and split training and test data by record, using only relational pair features.

On real ICD-10 identifiers, exact matching scores 0.000 on drift. Our learned verifier recovers those joins at 0.746, raising overall recall from 0.64 to 0.83 (+0.18, paired bootstrap $p < 10^{-3}$). A prefix-hierarchy rule ties the verifier (0.826 versus 0.826 overall, p=0.59), and canonicalization leaves exact matching at 0.000 on drift, ruling out a formatting cause. Jaro–Winkler (Winkler, 1990) ties the prefix rule at 0.729 on drift, while character term-frequency–inverse-document-frequency (TF-IDF) similarity (Sung et al., 2020) reaches 0.407. A structural rule is sufficient for these regular hierarchies. OCR/typo corruption weakens prefix matching (0.73 → 0.40), but our verifier does not significantly beat the best baseline in this two-way mix (0.58 versus 0.59, p=0.86). Five-seed results show the same pattern ($0.759_{\pm 0.027}$ versus prefix $0.791_{\pm 0.044}$). Consolidated baselines and corruption sweeps appear in the supplementary material.

Cross-vocabulary.

Semantic similarity becomes useful when the codes belong to different vocabularies. We join ICD-9 records to the ICD-10 KG over 322 diseases with 0% prefix overlap. Exact, fuzzy, and prefix matching all score ≈0.01, while embedding reaches 0.91 and our verifier 0.90.

Heterogeneous drift.

A clinical column can mix hierarchy-coded, OCR-corrupted, and legacy ICD-9 codes. Drawing one available mechanism per record gives a mixture on which prefix matching reaches 0.258 and embedding 0.689, while our learned verifier reaches 0.752 (+0.49 over prefix, $p < 10^{-3}$ in all five seeds, and +0.06 over embedding, significant in 3 of 5). The useful signal varies across records, favoring a combination of cues. Since our label audit found no-op OCR attempts whose relabeling changes only subtype metadata, we report aggregate performance without subtype-specific comparisons.

Comparison with tuned fusion.

Tuning cascade order and thresholds on the training split selects prefix matching first, followed by fuzzy matching at ≥ 0.9 and then embedding.

Tuned fusion can match learned verification. The cascade reaches $0.776_{\pm 0.021}$ and matches our verifier (difference -0.019, not significant in 4 of 5 seeds), as does logistic regression on the same six features.

Pool-relative features, defined as each score minus the maximum score in the candidate pool, halve our verifier's seed variance without changing its mean. These features expose the relative evidence that a cascade uses. The tested fusions reach 0.76–0.78. Our probabilistic verifier supplies the scores and uncertainty needed for abstention (Section 6.2), a use for which we have not calibrated the cascade.

6.2 End-to-end QA on the real table+KG benchmark

The component study leads to an end-to-end test on the same real identifiers and Wikidata relations. Using constructed clinical records under heterogeneous drift and the five question families from Section 4, we evaluate 518 questions per seed over five seeds. Only the gate changes between rows of Table 4. We report silent wrong answers (wrong) and abstention on answerable questions (false abst.). Utility U_λ scores a correct answer +1, a wrong answer $-\lambda$, and abstention 0. EM also rewards a gold abstention. In the table, τ is the embedding score threshold and UB denotes the oracle upper bound.

Table 4: End-to-end QA over constructed records with real ICD-10 codes and Wikidata relations: five seeds, identical planner, different join gates.

Gate	EM ↑	wrong ↓	false abst. ↓	$U_{\lambda=1}$	$U_{\lambda=2}$	$U_{\lambda=5}$
Exact code match	$0.354_{\pm 0.007}$	0.001	0.645	0.354	0.353	0.351
Prefix / hierarchy rule	$0.493_{\pm 0.014}$	0.003	0.505	0.490	0.487	0.479
Embedding cosine (τ=0.62)	$0.885_{\pm 0.021}$	0.111	0.004	0.775	0.664	0.332
No verifier	$0.860_{\pm 0.019}$	0.140	0.000	0.720	0.579	0.159
Learned verifier	$0.839_{\pm 0.015}$	**0.007**	0.154	0.832	0.825	0.804
Oracle (UB)	$0.935_{\pm 0.021}$	0.000	0.065	0.935	0.935	0.935

Gating reduces silent wrong answers on table+KG QA. Our learned gate lowers the wrong-answer rate from 14.0% to 0.7% over five seeds. EM is 0.839, within 2.1 points of ungated with no significant difference in any seed. Some answerable cases are deferred in exchange for fewer errors.

Exact and prefix gates abstain on half or more of the questions because their recall collapses on OCR and cross-vocabulary records. The ungated planner and an embedding gate reach the highest EM (0.860 and 0.885), but return silent wrong answers on 14.0% and 11.1% of questions and score 0.0 and 0.25 on the abstain family. Our learned gate's committed joins are 99.1% correct (system-level AURC 0.004), and our gate beats exact and prefix gates in 5/5 seeds. At every evaluated penalty $\lambda \in \{1, 2, 5\}$, our gated framework has the highest utility among non-oracle gates (0.804 versus 0.159 ungated at λ=5). False abstentions on hard OCR and sibling-code joins account for the remaining gap to the oracle (0.935).

6.3 BIRD value linking

BIRD (J. Li et al., 2023) is a text-to-SQL benchmark over 95 real databases with frequent value drift, such as "high school" in a question matching `HS` in a column. After resolving SQL aliases before identifying the filtered table, we extract 1,103 value-linking instances from the 1,534 development questions, of which 10.3% are drifted. We train our MLP ensemble on one split and evaluate on a held-out split, using identifier features and cosine similarity from the `bge-small` sentence-embedding model.

Across five corrected runs, exact matching reaches 0.808 ± 0.012 overall and 0.000 on drift. Fuzzy matching reaches 0.167 ± 0.038 on drift and embedding 0.806 ± 0.031.

Learned matching recovers drifted BIRD values. Across five corrected runs, our verifier reaches recall@1 0.939 ± 0.010 overall and 0.905 ± 0.051 on drift, where exact matching scores 0.000. Pair-level AUROC is 0.996 ± 0.002.

Correcting the candidate domains changes individual baseline scores, while the recovered drifted matches remain the main source of gain. Because the submitted cross-encoder run used the original extractor, we retain it in the supplement as an exploratory capacity check and exclude it from the corrected headline evidence.

6.4 End-to-end execution accuracy

We run text-to-SQL with a local Qwen2.5-Coder-7B-Instruct using 4-bit weights and score EX on BIRD development data. Our base pipeline reaches EX 0.458 on 800 questions (simple 0.53, moderate 0.35, challenging 0.37). Adding value hints without focusing the schema lowers EX to 0.446, motivating schema linking (H. Li et al., 2023, 2024; Wang et al., 2025). With a learned table selector at recall 0.93, full-development EX rises $0.454 \rightarrow 0.477$ over 1,534 questions, including 131 drifted cases (+2.3, p=0.008, CI $[+0.5, +4.2]$). Drift EX rises $0.33 \rightarrow 0.42$ (+9.2, p=0.005), and execution errors fall $0.164 \rightarrow 0.143$ under paired bootstrap with 10,000 resamples. A 14B coder shows the same +2.3 gain on 400 questions. Embedding-only pruning lowers EX (0.473 versus 0.476 with the full schema), learned pruning raises it to 0.483, and oracle pruning reaches 0.525.

Isolating the verifier end to end.

We fix schema pruning, column values, candidate pools, the column-relevance gate, and the three-hint prompt slot, varying only the value linker (Table 5). Two-fold cross-fitting scores each question with a verifier trained on the other fold. We use

an activation-aware weight-quantized (AWQ) 4-bit coder on NVIDIA CUDA, versus Apple's MLX in the submission. Baseline EX reproduces at 0.460 versus 0.454, with drift EX 0.344 versus 0.328. These generations retain the original schema-order extractor. Alias correction changes the training pools, and corrected generations have not been rerun. EX remains scored against BIRD gold SQL.

Table 5: Value-linker ablation on full BIRD dev (1,534 questions, 131 drifted), with schema pruning and column values fixed. Generations predate the alias correction.

Value linker	EX	EX drift	hinted q (prec.)	ΔEX [95% CI]	ΔEX, drift [95% CI]
no pruning, no values	0.460	0.344	0	-0.5 [-2.4,1.4]	-5.3 [-13.0,2.3]
none (pruning + values)	0.465	0.397	0	—	—
exact	0.406	0.328	1210 (0.14)	-5.9 [-7.6,-4.2]	-6.9 [-13.0,-1.5]
fuzzy	0.404	0.321	1412 (0.12)	-6.1 [-7.9,-4.2]	-7.6 [-13.0,-3.0]
embedding	0.451	0.359	808 (0.26)	-1.4 [-2.8,-0.1]	-3.8 [-10.7,2.3]
verifier, τ=0.5	0.460	0.359	549 (0.54)	-0.5 [-1.5,0.5]	-3.8 [-8.4,0.0]
verifier, τ=0.8	0.469	0.405	321 (0.63)	+0.4 [-0.3,1.1]	+0.8 [-3.8,5.3]
oracle gold literals	0.495	0.496	828 (0.98)	**+3.1 [1.9,4.2]**	**+9.9 [4.6,16.0]**
oracle, drifted only	0.469	0.458	107 (1.00)	+0.5 [0.0,0.9]	+6.1 [0.8,11.5]

In Table 5, "hinted q" counts questions receiving hints and parentheses give hint precision. The threshold τ controls which hints we supply. ΔEX is the percentage-point change from no hints, with paired-bootstrap 95% CIs from 10^4 resamples. Intervals crossing zero include both gains and losses.

Attributing the gain.

Gold literals in the same advisory slot raise EX by +3.1 overall and +9.9 on drift, and drifted literals alone raise drift EX by 6.1 points. Our verifier is the best non-oracle linker in this run, with hint precision 0.54–0.70, four times that of exact or fuzzy linking. Those linkers lower EX by about six points.

Accurate value matching alone yields limited SQL gains. Verifier hints change EX by +0.4 points at τ=0.8 in the original-extractor run, with a confidence interval spanning zero. Column selection and identifying value mentions remain necessary.

Our verifier confidence helps select useful hints, as EX on hinted questions rises from 0.495 to 0.577 when we defer the least-confident half. Column grounding accounts for part of the remaining oracle gap, even though our corrected verifier reaches five-seed recall@1 0.91 on drifted values in a known column. The pruning-plus-values gain also depends on the backend, changing from +2.3 EX with MLX (p=0.008) to 0.465 versus 0.460 with AWQ (+0.5, not significant). The supplementary material provides the schema-linker and backend comparisons.

6.5 Negative result and limits

In HybridQA (Chen et al., 2020), gold cell-to-passage links remove the identity-linking step. We must retrieve the passage containing the answer, and lexical overlap beats dense retrieval (recall@20 0.83 versus 0.65).

Lexical retrieval remains stronger on HybridQA. With identity links supplied, our identifier-trained verifier reaches top-1 recall 0.09 versus 0.25 for lexical overlap. Stronger retrievers and a fine-tuned cross-encoder (0.21) leave this gap.

A cheap-first cascade matches lexical HybridQA recall (0.25) but regresses on BIRD. Verifier-first preserves BIRD and partially recovers HybridQA (0.15). Separately trained scores have incompatible scales, with BIRD-verifier ECE 0.117 on HybridQA. Training jointly on pooled pairs with common, pool-relative, and input-type features improves calibration and raises HybridQA top-1 from $0.146_{\pm 0.043}$ to $0.237_{\pm 0.030}$, closing two thirds of the lexical gap ($0.281_{\pm 0.030}$, three seeds). This E4 run predates alias correction and provides exploratory routing evidence. Retrieval curves and cascade comparisons appear in the supplement.

7 Discussion

The effect of a match score depends on the decision it controls. When our table+KG planner commits or abstains on the verifier's judgment, silent wrong answers fall by 13 percentage points (Section 6.2). The same evidence supplied as an advisory SQL hint adds a dependency, because the coder must first identify the relevant column, and only a strict threshold yields a small gain (Table 5).

To quantify this choice we measure the inputs to Theorems 1 and 2 on held-out pairs. The critical penalty is $\lambda^\star = 0.39$ on synthetic pairs and 0.0013 on alias-corrected BIRD seed-0 pairs, which predicts a utility gain at every evaluated $\lambda \in \{1, 2, 5\}$, and the predicted sign matches measured synthetic QA utility in every seed. The corollary overstates the reduction in wrong answers (observed QA wrong-rate ratio $0.77_{\pm 0.07}$ versus $\alpha = 0.33$) because QA-time distractors are real neighboring nodes, harder than the pairs used to measure α, whereas Theorem 2 predicts the controlled recall gain closely (predicted $0.512_{\pm 0.032}$, observed $0.515_{\pm 0.025}$). The supplementary material lists the measured inputs, latency, noise robustness, and a Fellegi–Sunter baseline (AUROC 0.753 versus our 0.820).

The remaining errors point to different remedies. Removing 10% to 30% of KG edges lowers EM from 0.94 to 0.52 while the wrong-answer rate stays under 0.05, because the planner abstains on missing evidence. Of the 41 recall@1 misses on ICD-10, 18 confuse sibling codes sharing a category, and we commit 38 of the 41 wrong top-ranked candidates, which calls for better discrimination or uncertainty estimates. Gating also rejects 6% of correct positives on clean synthetic joins (EM $1.00 \rightarrow 0.94$), and our identifier-trained verifier underperforms lexical matching on HybridQA. Retrieval that helps only behind a validation gate also appears in LLM program repair (X-B. Le et al., 2026). We keep direct matching for clean identifiers and lexical retrieval for passage selection, and reserve the learned gate for drifted joins.

Limitations and Future Work

Our end-to-end text-to-SQL evidence comes from a local 7B coder at EX 0.45–0.48, with gains that depend on the backend and on column grounding (Table 5), and the expensive BIRD generations use the initial extractor while our corrected reruns cover component matching. For table+KG QA we use constructed clinical records and a

rule-guided planner over a 545-node KG, and the synthetic setting favors the ungated baseline by usually including the gold node in retrieval. We join one table to one graph and test composite keys (AUROC 0.93/0.89 for AND/OR composition). Multi-table joins, aggregation, set operations, and temporal drift lie outside this pairwise verifier, synthetic drift may not represent real key formats, and learned planning, larger graphs, and further clinical terminologies would broaden the evaluation.

8 Conclusion

We find learned join verification most useful when a planner can reject candidates or abstain directly on its verdict. Across our experiments, we recover drifted identifiers and reduce silent wrong answers by allowing the planner to abstain. Advisory SQL hints have a smaller effect because column selection and query generation introduce further decisions. Before choosing a gate, we recommend measuring verifier errors, retrieval failures, and the cost of a wrong answer. Our conditional theory and released benchmarks support this assessment alongside alternatives such as hierarchy rules, tuned cascades, and lexical matching.

Acknowledgments

We acknowledge Ho Chi Minh City University of Technology (HCMUT), VNU-HCM for supporting this study.

Statements and Declarations

- **Funding:** Not applicable.
- **Competing interests:** The authors declare no competing interests.
- **Ethics approval, consent to participate, and consent for publication:** Not applicable.

- **Data and code availability:** BIRD, HybridQA, ICD-10-CM, and Wikidata are public. The core harness, constructed ICD-10 + Wikidata benchmarks, pinned splits, and original experiment outputs are available under the MIT license at https://github.com/lxbachcs/relationalrag-tables. The repository also contains the completed annotation audit, the alias-correction driver, and the corrected snapshot used for the corrected BIRD rows. An archived, DOI-bearing release will accompany the camera-ready version.
- **Declaration of generative AI use:** The authors used AI assistants (Claude, Anthropic; Codex, OpenAI) to help write experiment drivers and edit the text. The authors reviewed and edited all content, ran and checked every experiment themselves, and take full responsibility for the publication. No generative AI was used to produce data, results, or figures.
- **Author contributions:** All authors contributed to the study conception and design. X.-B. Le developed the system, ran experiments, and drafted the manuscript. P.T. Tran Truong contributed to data curation and evaluation scripts. H.X. Son supervised the project and revised the manuscript. All authors read and approved the final manuscript.

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